

PRODDUTURI SHARATH CHANDRA

AI ENGINEER | GENERATIVE AI | LLM & MULTI-AGENT SYSTEMS | RAG PIPELINES

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PROFESSIONAL SUMMARY

AI Engineer with experience in architecting and deploying Generative AI ecosystems, specializing in multi-agent orchestration, advanced RAG pipelines, and instruction fine-tuning for Small Language Models (SLMs). Proficient in building highly scalable, fault-tolerant AI backend infrastructure using Python, LangChain, FastAPI, and vector databases. Strong focus on bridging robust full-stack engineering with LLM capabilities to deliver low-latency enterprise applications.

TECHNICAL SKILLS

Generative AI:	RAG Architecture, Multi-Agent Systems, LangChain, LlamaIndex, Prompt Engineering, Function/Tool Calling, Fine-Tuning (QLoRA), Gemini / Llama Integration
ML & Data & Storage:	PyTorch, HuggingFace, Pinecone (Vector DB), Hybrid Search (Dense + Sparse, BM25), PostgreSQL (Relational & JSONB), MongoDB
Languages:	Python, JavaScript / TypeScript, Java, SQL
Backend & Infrastructure:	FastAPI, Node.js, Express.js, React.js, GraphQL, Docker, vLLM, Unsloth, Git, Cursor, GitHub Copilot

PROFESSIONAL EXPERIENCE

LTIMindtree

Software Engineer

June 2025 – Present

Bangalore, India

- **Collaborated** in developing full-stack enterprise Self-Service Portal (SSP) using React, Node.js, Express, and SQL, serving as the centralized hub for internal automation and enterprise data orchestration.
- **Integrated an LLM-driven conversational agent** into the portal, implementing dynamic intent recognition and function/tool calling to parse natural-language user requests in real time.
- **Engineered the backend** to translate parsed intents into strictly validated, structured JSON API payloads, integrating seamlessly with distributed enterprise systems.
- **Developed automated data pipelines** to orchestrate complex, multi-step API transactions spanning authentication, dynamic rate querying, and secure tokenization, reducing manual data provisioning time by 70%.
- **Accelerated the platform development lifecycle** by leveraging AI-assisted programming tools (Cursor, Copilot) for rapid full-stack development.

HCL Technologies

Software Engineering Intern

Feb 2024 – May 2024

Chennai, India

- **Contributed** to the development of full-stack web applications & scalable RESTful architectures within an Agile environment.

CORE AI ENGINEERING PROJECTS

Enterprise Multi-Agent AI Orchestration Platform

Python, LangChain, FastAPI, React, PostgreSQL, Docker

- Engineered an asynchronous **multi-agent orchestration engine** that translates natural-language queries into parallelized **DAGs**, reducing overall task execution latency by 40%.
- Designed an intelligent LLM routing dispatcher that enforces strict **schema validation**, guaranteeing **deterministic, structured outputs** and eliminating malformed API payloads.
- Implemented a self-healing **LLM invocation pipeline** featuring cross-provider **fallback routing** and exponential backoff, achieving **100% system uptime** against external API rate limits.
- Parallelized sub-task execution using Python's **asyncio**, backed by a hybrid **PostgreSQL** state management system to persist complex, unpredictable LLM workflows.

SLM Instruction Fine-Tuning

Python, PyTorch, Unsloth, QLoRA, HuggingFace TRL, vLLM

- Fine-tuned an open-weights **Llama 3 (8B)** model using **Unsloth** and **QLoRA**, optimizing the model to extract strictly typed **JSON payloads** (issue severity, component, intent) from unstructured enterprise support tickets.
- Curated a synthetic training dataset of 5,000+ support interactions and evaluated model performance based on **JSON Schema Validation Pass Rate**, achieving a 98% perfectly parsable output rate.
- Implemented **Parameter-Efficient Fine-Tuning (PEFT)** to update <2% of total model parameters, drastically reducing VRAM requirements for training while preventing catastrophic forgetting.

Enterprise RAG Document Intelligence Pipeline

Python, LangChain, FastAPI, Pinecone

- Engineered a scalable **Retrieval-Augmented Generation (RAG)** pipeline to ingest, chunk, and embed large-scale proprietary text datasets into a **Pinecone vector database** for low-latency retrieval.
- **Implemented a hybrid search architecture** fusing dense vector embeddings with sparse keyword search (BM25), improving context retrieval accuracy by **35%**.
- Designed asynchronous **FastAPI** endpoints to orchestrate contextual querying via **LangChain**, dynamically injecting retrieved vector chunks to reduce model hallucinations by over **85%**.

EDUCATION & PUBLICATIONS

Sathyabama Institute of Science and Technology, Chennai, Tamil Nadu

2021 – 2025

Bachelor of Engineering in Computer Science • CGPA: 8.13 / 10.0

Publication: "Diagnosis of Neurodegenerative Diseases Using Deep Learning Methods" — Research Presented at ICRETM 2025.